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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE

EXAMINATION, APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) MG2DSCMBG101 MICROBIAL PHYSIOLOGY AND FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A- MCQ Answer all Questions (1mark each)

1. a) Glucose
2. c) To indicate changes in pH
3. a) Liquid medium
4. d) Ammonia
5. b) Rhizopus stolonifer
6. c) Fatty acids and glycerol
7. a) Spoilage
8. d) All of the above
9. c) Sulfur dioxide
10. d) All of the above

Part B – Fill in the blanks Answer all Questions (1mark each)

11. Log phase
12. MacConkey agar
13. Nitrogen
14. Perishable foods
15. Quick freezing

Part C – State True/False Answer all Questions (1mark each)

16. False
17. False
18. True
19. True
20. False

Part D – Very Short Answers Answer any 10 Questions(1mark each)

21. Energy from inorganic compounds (1mark)
22. Potato Dextrose Agar / medium for growth of fungus (1mark)
23. Require oxygen(1mark)
24. Nitrogen source(1mark)
25. High temperature loving microbes(1mark)
26. Skin, hair, droplets (any two-1/2 mark each)
27. Yeast(1mark)
28. Benzoic acid , lysozyme etc (any two-1/2 mark each)
29. UV, Infrared (any two-1/2 mark each)
30. Heat + anaerobic condition (1/2 mark each)
31. Phenols, formaldehyde (any two 1/2 mark each)
32. Generally Recognized As Safe(1mark)

Part E – Short Essay Answers. Answer any 4 (3 marks each)

33. Compare batch culture and continuous culture

Batch culture: Closed system; nutrients added once; shows all growth phases. (1.5marks)
Continuous culture: Open system; fresh medium added continuously; cells remain in log phase.
Used for industrial production.(1.5marks)

34. Explain spread plate method
A diluted sample is spread on agar using a sterile spreader. After incubation, colonies are counted.(2marks)
Used for isolation and enumeration of microbes.(1mark)

35. Significance of bacteria in food industry
Lactobacillus – curd/yogurt production.(1mark)

Acetobacter – vinegar production.(1mark)

Streptococcus thermophilus – dairy fermentation.(1mark)

36. Sources of contamination from animals
Feces, skin, hair, feathers, infected milk, slaughtering process, pathogens like Salmonella. (3marks)

37. Pasteurization and significance
Heat treatment (e.g., 72°C for 15 sec) to kill pathogens.(1.5marks)
Extends shelf life and ensures food safety(1.5marks)
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38. Antibiotics as food preservatives
Used to inhibit microbes (e.g., nisin in dairy).(1.5mark)
Extends shelf life but limited due to resistance concerns.(1.5mark)

Part F – Long Essay Answers. Answer any 1 (8 marks each)

39. Selective medium to inhibit Gram-positive bacteria
Use bile salts and crystal violet to inhibit Gram-positive bacteria.(3marks)
Example composition: Peptone, lactose, neutral red, bile salts, crystal violet.(3marks)
Supports Gram-negative growth.(2marks)

40. Low temperature preservation & differences
Low temperature slows enzymes and microbial growth. (2marks)
Chilling: 0–5°C, short-term storage, microbes slowed.(2marks)
Freezing: below −18°C, long-term storage, microbes almost stopped.(2marks)
Freezing forms ice crystals and may affect texture.(2marks)